

BACKEND DISCUSSION GUIDE

AI Pod Staffing

From Excel-backed prototype to governed production platform

Simple architecture, data flow, agent flow and decisions required

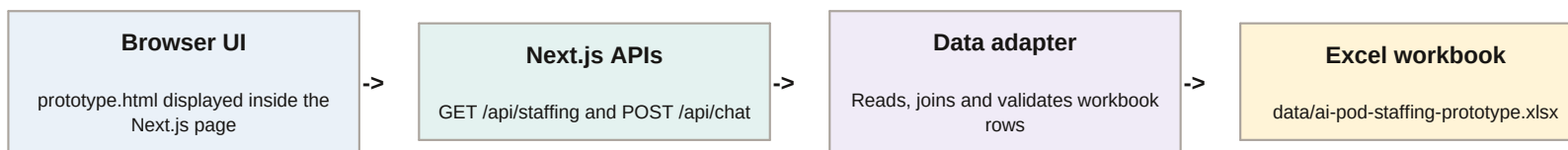
Meeting use

Use this after the frontend walkthrough

Current state is shown separately from the proposed target state

1. What exists today

A small Next.js application reads the workbook and serves the frontend prototype.



Current runtime flow

- The browser opens the prototype and calls the staffing API.
- The server reads the Excel workbook from the data folder.
- The adapter combines project types, deliverables, skills, people, availability, requests and recommendations.
- Reference checks stop the load if important IDs do not match.
- The API returns one UI-ready JSON response and the browser renders it.
- The workbook is read again when the file modification time changes.

What the two APIs do

API	Purpose today	Production gap
GET /api/staffing	Returns workbook-backed catalogue, people, requests, availability, recommendations and metrics.	Read-only. No create, update or decision persistence.
POST /api/chat	Returns role-scoped answers using keyword and rule-based prototype logic.	No LLM, agent runtime, retrieval service or conversation audit yet.

CURRENT LIMITATION	Saving a request, availability event or approval does not update Excel or a database. Browser actions are demonstration-only.
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2. Excel workbook as the current data model

The workbook provides catalogue, people, demand, capacity and recommendation evidence.

Customer catalogue and mapping

Sheet	Rows	What it represents
Project Types	11	Customer project categories and descriptions.
Deliverables	55	Customer deliverables linked to project types.
Deliverable Skills	87	Default capability mappings for each deliverable.
Interests	13	Controlled skill or type-of-work catalogue.
Customer Mapping	67	Traceability to the supplied customer source mapping.

Operational staffing data

Sheet	Rows	What it represents
People	8	People, role, location, allocation percentage and active pod count.
Person Interests	27	Person-to-capability strength and supporting evidence.
Availability	4	Leave, travel and other recorded capacity events.
Requests	3	Staffing demand, dates, priority, owner and status.
Requirements	10	Required capabilities for each request and their source.
Recommendations	3	Proposed people, pod role, score, rationale and decision status.

TRACEABILITY

Source version identifies the customer mapping version. Source row identifies the original row used to create a project or deliverable, helping reviewers trace displayed content back to the supplied workbook.

DATA OWNERSHIP

Excel is currently a demonstration source, not a safe multi-user transaction system. The customer must confirm the future system of record and the owner of each data domain.

3. Proposed production backend

Replace the workbook runtime with governed services and Oracle Database 26ai while retaining the current frontend experience.

Recommended logical layers

Layer	Responsibility
Experience	Web application, role-specific workspaces and conversational Ask AI Pod.
Access	Enterprise SSO, API gateway, authorization and persona/data-scope enforcement.
Business services	Request, catalogue, people, capacity, recommendation, decision, assignment, notification and reporting services.
Agent orchestration	AIDP or enterprise agent orchestration coordinating tools, evidence gathering, rationale and the human checkpoint.
Deterministic tools	Eligibility, capacity, conflict, pod-size, scoring and policy checks implemented as testable services.
Data	Oracle Database 26ai for operational records, audit, JSON evidence and vector search where useful.
Enterprise integrations	Identity, HR/people, project work, calendars/leave, skills, notification and reporting sources.

Simple target flow



IMPORTANT DESIGN RULE

Use agents for orchestration, explanation and flexible reasoning. Use deterministic services for eligibility, dates, allocation limits, conflicts, authorization and final state changes.

CURRENT SQL NOTE

The repository contains a useful draft relational model, but it uses PostgreSQL-style syntax. It must be converted and reviewed as Oracle-native database design before Oracle Database 26ai deployment.

4. Agentic recommendation and human-in-the-loop flow

The agent prepares a recommendation; a responsible person makes the staffing decision.

Step	Stage	Backend responsibility
1	Capture request	Store project, deliverables, capabilities, dates, effort, pod size and business outcomes.
2	Validate	Confirm required fields and customer-controlled taxonomy.
3	Gather evidence	Retrieve people, capability evidence, delivery history, assignments, leave and travel.
4	Apply hard rules	Exclude unavailable or ineligible people and enforce mandatory capabilities and access scope.
5	Rank candidates	Calculate fit factors and preserve factor-level evidence.
6	Explain	Generate a clear rationale, alternatives, exclusions and capacity warnings.
7	Human decision	Approve, adjust or decline. Require a reason when the proposal is changed or declined.
8	Finalize	Persist decision, create assignments, notify people and write the audit record.

Human decision states

Decision	What the backend records
Approve	Selected pod, approver, timestamp, comment and evidence version.
Adjust	Original proposal, changed members or roles, override reason, approver and timestamp.
Decline	Decline reason, approver, timestamp and whether a re-run is requested.

CURRENT UI GAP

The prototype currently demonstrates Approve Pod. Adjust and Decline controls, mandatory reasons and database-backed audit updates still need to be implemented.

5. Core backend services and APIs

Keep responsibilities clear so the UI, agents and integrations use the same governed rules.

Service	Example operations	Why it exists
Catalogue	Read project types, deliverables and capabilities; publish approved mapping versions.	Maintains customer terminology and traceability.
Staffing requests	Create, update, search, close and retrieve requests.	Owns demand and its lifecycle.
People and skills	Retrieve profiles, roles, capability evidence and relevant history.	Provides candidate evidence within access scope.
Capacity	Retrieve assignments, leave, travel and calculated allocation.	Creates one trusted capacity view.
Recommendation	Run fitment, retrieve recommendations, alternatives, exclusions and evidence.	Coordinates agent and deterministic tools.
Decision and assignment	Approve, adjust, decline and create assignments.	Enforces the human checkpoint and final state changes.
Audit	Read immutable recommendation, decision and configuration history.	Supports governance and review.
Reporting	Operational KPIs and role-scoped summaries.	Keeps calculations consistent across screens.
Assistant	Role-scoped conversational queries with citations to source records.	Provides governed natural-language access.

Recommended API behavior

- Every write validates the user's role and data scope on the server.
- Every recommendation stores the exact input snapshot, tool versions and policy version.
- Repeated requests use idempotency controls so retries do not create duplicate assignments.
- Long-running fitment jobs expose a status and correlation ID rather than blocking the browser.
- Errors are business-readable and do not silently fall back to invented data.

6. External data required

Recommendations are only reliable when the customer confirms the authoritative source for each input.

Data needed	Likely source	Decision required
Identity, role and persona	Enterprise identity and HR	Who assigns Operations Lead, Request Lead, Pod Lead and POD Member roles?
People and organization	HR or resource management	Which people are eligible for this staffing process?
Capability catalogue	Customer catalogue or skills platform	Who owns capability names and approves changes?
Person capability evidence	Skills profile, manager validation and delivery history	Can members self-declare? Who validates strength and evidence?
Allocation percentage	Project or resource management system	What work counts and how is percentage calculated?
Reserved operational capacity	Business policy	How much capacity is protected from project staffing?
Leave, travel and availability	Calendar, leave and travel systems	Which events block staffing and how current must they be?
Project assignments and history	Project/work management	What defines an active pod and completed delivery evidence?
Approvals and notifications	Workflow and messaging	Who approves each request and where should notifications be sent?

MINIMUM FOR THE FIRST PRODUCTION RELEASE

Identity, people, approved capability catalogue, current assignments, availability, staffing requests, recommendation evidence, human decisions and an immutable audit trail.

7. Security, governance and operations

Production authorization and audit must be enforced in the backend, not only hidden in the frontend.

Control	Required behavior
Authentication	Enterprise SSO identifies the user and service accounts.
Authorization	Role and row-level rules restrict requests, people, reports and actions.
Data protection	Encrypt data in transit and at rest; restrict sensitive profile and availability fields.
Human approval	No assignment is finalized without an authorized recorded decision.
Auditability	Store input snapshot, factors, exclusions, rationale, model/tool versions, decision and overrides.
AI governance	Version prompts and policies, evaluate recommendation quality and prevent unsupported answers.
Observability	Track correlation IDs, job status, latency, failures, retries and business outcomes.
Data quality	Reject broken references and surface stale or missing source data.
Retention	Agree retention and deletion rules for prompts, recommendations, decisions and employee data.

Failure behavior to agree

- If a source system is unavailable, show that evidence is incomplete and do not fabricate availability or allocation.
- If a recommendation tool fails, keep the request open and allow a controlled retry.
- If the final assignment write fails, do not mark the request as staffed.
- If authorization cannot be confirmed, deny access and record the attempt.

8. Practical implementation phases

Build the reliable data and decision foundation before adding greater agent autonomy.

Phase	Outcome
0. Confirm rules and ownership	Agree personas, source systems, calculations, guardrails, approvals and data owners.
1. Controlled Excel ingestion	Load the workbook into staging, validate references and publish an import report. Keep Excel only as a migration input.
2. Core Oracle data and APIs	Implement catalogue, people, request, capacity and role-scoped read/write services in Oracle Database 26ai.
3. Deterministic fitment tools	Implement eligibility, conflicts, capacity and scoring as versioned testable services.
4. Human decision workflow	Implement approve, adjust and decline with reasons, assignment creation, notifications and immutable audit.
5. Agent orchestration	Add governed evidence gathering, rationale, alternatives and Ask AI Pod with role-scoped retrieval.
6. Enterprise integrations	Connect authoritative identity, HR, skills, assignments, calendar, leave, travel and reporting systems.
7. Pilot and hardening	Run evaluation, security, performance and user acceptance testing; monitor recommendation quality and overrides.

Suggested meeting wording

SAY

Today the workbook gives us a controlled way to demonstrate the experience and validate the customer's terminology. For production, Excel becomes a migration and reference input, while Oracle Database and governed services become the system of record. The agent will coordinate evidence and explain recommendations, but deterministic rules will enforce eligibility and capacity, and a human will approve, adjust or decline every final staffing decision.

9. Questions to resolve with the customer

Use simple business language and identify the owner of each answer.

1. Where should the official list of people and their roles come from?
2. How is allocation percentage calculated, and how often is it refreshed?
3. How much operational capacity must remain protected from project staffing?
4. How are Pod Lead and POD Member roles assigned and changed?
5. Can POD Members declare their own capabilities? Who validates the strength and evidence?
6. What makes someone eligible or ineligible for a request?
7. Which leave, travel, calendar or project commitments must block a recommendation?
8. Who is allowed to approve, adjust or decline a proposed pod?
9. When someone adjusts or declines, which reasons must be captured?
10. What request and pod statuses are needed from creation through closure?
11. Which system should receive the final staffing assignment?
12. Who should be notified when a recommendation is ready or a pod is approved?
13. Which historical delivery evidence can the solution use?
14. How long should recommendation evidence, decisions and employee data be retained?
15. Which reports and KPIs are required for Operations Leads and executives?

MEETING OBJECTIVE

Leave the discussion with named owners for data, rules and approvals, plus agreement on the first production scope. Do not attempt to finalize technical products before these business decisions are clear.